Potential Research Gaps

Version 2 | Narrowed Research Directions

Following feedback on Version 1, the original action-selection and speech-specific ambiguity gaps have been consolidated. Automatic speech recognition (ASR) errors and acoustic noise now serve as the experimental setting for studying voice-agent decisions. Three distinct, narrower research candidates are proposed below.

Gap 1 — Noise-Marginalized Value of Information (NM-VoI)

Unsolved problem: Current decision frameworks, such as expected value of perfect information (EVPI), can assume that asking a clarification question yields a perfectly transcribed response. This assumption may fail in noisy voice interactions, where the reply itself can be misrecognized.

Proposed contribution: Develop a dynamic clarification policy that accounts for acoustic noise, represented by signal-to-noise ratio (SNR), when estimating the value and cost of asking. When the expected reply is unlikely to be understood, the agent may abstain instead of entering repeated clarification loops.

Proposed experiment: Benchmark against a standard EVPI-based policy using the SLURP dataset with synthetic acoustic noise. Compare task success and number of conversational turns across SNR levels.

Research question: How should a voice agent adjust the expected value of clarification under noisy acoustic conditions when choosing whether to answer, clarify, or abstain?

Gap 2 — Joint Acoustic-Semantic Source Attribution (JASSA)

Unsolved problem: Post-hoc language-model prompting for error diagnosis may overlook raw acoustic evidence, making it difficult to distinguish speech-recognition errors from genuine semantic ambiguity.

Proposed contribution: Fuse ASR lattice confidence and language-model token distributions before the decision layer to attribute uncertainty to acoustic recognition errors or semantic ambiguity.

Proposed experiment: Evaluate uncertainty-source classification using F1 and AUROC on the Audio MultiChallenge dataset, subject to confirming the availability of suitable acoustic evidence and ground-truth attribution labels.

Research question: Can joint acoustic and semantic evidence more accurately identify the source of uncertainty than post-hoc text-only diagnosis?

Gap 3 — Multi-Turn Acoustic Calibration (Audio-ECE)

Unsolved problem: Existing multi-turn calibration measures, such as ECE@T, may not capture the accumulation of ASR hallucinations and recognition errors across dependent spoken turns.

Proposed contribution: Investigate an SNR-aware decay factor for belief updates and develop an evaluation measure for multi-turn voice calibration.

Proposed experiment: Study changes in Brier score across dependent, noisy conversational turns, comparing acoustic conditions and turn depth.

Research question: How can calibration be measured and maintained when recognition uncertainty accumulates over multiple spoken turns?

Direction

NM-VoI is the preferred candidate because it directly connects acoustic uncertainty to the central action-selection decision: answer, clarify, or abstain. JASSA focuses on identifying the source of uncertainty, while Audio-ECE focuses on evaluating uncertainty across multiple turns. The final choice should follow a feasibility and novelty review of the proposed methods, datasets, and baselines.